SERGIO BARRIENTOS

AI Engineer | Linux Platforms, Local LLMs & Applied GenAI

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PROFESSIONAL SUMMARY

Linux platform engineer moving into applied AI, with enterprise experience in regulated banking and railway environments. I combine production Linux operations, automation, and reliable deployment practices with hands-on local-LLM engineering and shipped AI-enabled Android products. My current work includes internal Azure AI Foundry and Document Intelligence exploration for document-processing workflows; independently, I build and benchmark local inference systems across Intel, AMD, and RK3588 hardware. I bring an infrastructure-first perspective to building AI applications that can be operated and maintained.

WORK EXPERIENCE

Linux and Virtualization Engineer | Deutsche Pfandbriefbank AG

Jul 2025 - Present

Full-time | Hybrid | Madrid, Spain | BaFin/ECB-supervised bank | 300+ server estate

- **Linux platform operations:** Work within the infrastructure team supporting a 300+ server RHEL estate across distributed datacenters and banking-critical workloads.
- **RHEL modernization:** Lead the bank's RHEL 8 to 9 migration project, coordinating the phased technical work and using Red Hat Satellite for provisioning, software distribution, patching, and lifecycle management.
- **Automation:** Build and maintain Ansible playbooks and Git-based workflows for server deployment, controlled infrastructure changes, and reproducible configuration.
- **Reliability and security:** Support hardening and patching cycles, CheckMK monitoring, high-availability clusters, disaster-recovery activities, and VMware vSphere operations.
- **Application support:** Work with Application Owners on Linux requirements and troubleshoot services including Nginx, Apache, Tomcat, JBoss, and PostgreSQL.
- **Applied GenAI:** Contribute to internal exploration and prototypes using Azure AI Foundry and Document Intelligence for document-processing and automation use cases, with attention to controlled deployment and operational requirements.

IT Application Specialist / Linux System Administrator | Sopra Steria

Apr 2023 - Jul 2025

Internship followed by full-time role | Hybrid | Madrid, Spain | Norwegian railway-sector client | 100+ servers

- **Technical coordination:** Coordinated a five-person team supporting, maintaining, and patching a critical application environment spanning more than 100 Linux and Windows servers.
- **Linux support:** Progressed from an internship into a full-time application support role and later a Linux-focused assignment, troubleshooting Red Hat Linux, Windows, and Azure environments.
- **Automation:** Used Ansible to automate repeatable administration and deployment tasks and supported RHEL lifecycle activities with Red Hat Satellite.
- **Enterprise operations:** Managed incidents, requests, and controlled changes through ServiceNow within an ITIL-aligned support model.
- **Identity and collaboration:** Supported Active Directory, Kerberos, Entra ID, Intune, and Exchange while working in English with a Norwegian client.

TECHNICAL SKILLS

Linux / RHEL	RHEL 7/8/9, Red Hat Satellite, LVM, systemd, SELinux, Kickstart, HA/DR, performance tuning
Automation / IaC	Ansible, Bash scripting, Git-based workflows, repeatable infrastructure configuration
Applied AI / Local LLMs	llama.cpp, local inference, GGUF quantization, Intel SYCL, AMD ROCm, RK3588 edge inference, KV-cache tuning, concurrency testing, benchmarking
Platforms / Services	VMware vSphere/vCenter, Nginx, Apache HTTP Server, Tomcat, JBoss Web Server, PostgreSQL, SSL/TLS
Cloud / Applied GenAI	Microsoft Azure, Azure AI Foundry, Document Intelligence, LLM prototyping, Firebase, Git-based delivery workflows
Operations	CheckMK, ServiceNow, Active Directory, Kerberos integration, ITIL, change management

SELECTED PROJECTS

- **IntelliAuto:** Published Kotlin/Firebase vehicle-management app with ML Kit OCR, offline encrypted storage, maintenance workflows, and an AI assistant.
- **IntelliFlow:** Privacy-focused personal-finance app with offline-first storage, cloud sync, projections, and AI-generated spending summaries.
- **Engineering portfolio:** Reproducible local-LLM work across llama.cpp, model quantization, Intel SYCL, AMD ROCm, RK3588, KV-cache tuning, concurrency, and power efficiency.

CERTIFICATIONS & EDUCATION

RHCSA - Red Hat Certified System Administrator — Sep 2024

Technician in Computer Systems and Networks Administration — Jun 2023 | Madrid, Spain

LANGUAGES

Spanish: Native **English:** Professional working proficiency